WORKSHEET

Chapter 6: Simple and compound interest

Simple interest

$$I = \frac{Prn}{100}$$

Compound interest

$$A = P\left(1 + \frac{r}{100}\right)^n \quad \textbf{(final amount)}$$

$$I = A - P \quad \textbf{(interest)}$$

where P is the principal
r is the interest rate per period expressed as a percentage
n is the number of periods
Use 1 year = 52 weeks = 365 days.

1 Write as decimals.

a 4%

b 7.8%

c 10.3%

d 16%

e 20%

f 5.12%

2 How many months in:

a 4 years?

b 9 years?

c $5\frac{1}{2}$ years?

d $2\frac{1}{2}$ years?

3 Convert these rates to monthly interest rates.

a 9% p.a.

b 3.9% p.a.

c 11.7% p.a.

d 16.2% p.a.

4 Convert these rates to daily interest rates, correct to four decimal places.

a 17% p.a.

b 20.2% p.a.

c 8% p.a.

d 4.08% p.a.

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5 Calculate the simple interest earned from each of the following investments.

a $5600 at 7% p.a. for 3 years

b $18 100 at 10.2% p.a. for 5 years

c $7500 at 6.3% p.a. for $2\frac{1}{2}$ years

d $26 000 at 9.5% p.a. for 8 months

e $10 000 at 12% p.a. for 6 months

f $4900 at 8.1% p.a. for 100 days

g $3200 at 0.48% per month for 1 year

h $2790 at 0.0603% per day for 55 days

6 What amount must be invested at 5.7% p.a. for 3 years to earn $651.51 in simple interest?

7 Lim's investment of $2100 earned $404.25 in simple interest after $3\frac{1}{2}$ years. What was the interest rate p.a.?

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8 For how long must Kimberly invest $4130 at 3.75% p.a. for it to earn $929.25 in simple interest?

9 Calculate the final amount of each investment accumulating compound interest.

a $5600 at 7% p.a. for 3 years

b $18 100 at 10.2% p.a. for 5 years

c $7500 at 6.3% p.a. compounded half-yearly for $2\frac{1}{2}$ years

d $26 000 at 9.5% p.a. compounded monthly for 8 months

e $10 000 at 12% p.a. compounded monthly for 6 months

f $3200 at 0.48% per month for 1 year

10 $20 000 is invested at 4.75% p.a. compound interest for 3 years. Calculate:

a the final amount of the investment

b the total interest earned.

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11 $9850 is invested at 0.4167% compound interest per month for 10 months. Calculate:

a the final amount of the investment

b the total interest earned.

12 What is the principal that must be invested at compound interest to reach each of the following final amounts? Answer to the nearest cent.

a $8000 in 4 years at 5% p.a.

b $14 200 in 3 years at 7.2% p.a.

13 Calculate the compound interest earned when $13 500 is invested at 7% p.a. compounded quarterly for 3 years.

14 By guessing and checking, find how long it will take an investment of $5000 to grow to $8812.85 if the compound interest rate is 6.5% p.a. Answer correct to the nearest year.

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Answers

1 **a** 0.04
b 0.078
c 0.103
d 0.16
e 0.2
f 0.0512

2 **a** 48 months
b 108 months
c 66 months
d 30 months

3 **a** 0.75%
b 0.325%
c 0.975%
d 1.35%

4 **a** 0.0466%
b 0.0553%
c 0.0219%
d 0.0112%

5 **a** $1176
b $9231
c $1181.25
d $1646.67
e $600
f $108.74
g $184.32
h $92.53

6 $3810

7 5.5%

8 6 years

9 **a** $6860.24
b $29 416.20
c $8758.05
d $27 693.02
e $10 615.20
f $3389.26

10 **a** $22 987.52
b $2987.52

11 **a** $10 268.23
b $418.23

12 **a** $6581.62
b $11 526.67

13 $3124.43

14 9 years

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